

Meaning: 0/0 of Language (Symbols as Shared Prediction)

Michael Grafiel S Puno

2026

The Removable Singularity Framework

Abstract

After empathy-as-information (Ch.60), the machines that make "thou" legible at scale are SYMBOLS. A token has no intrinsic meaning: meaning is role (Wittgenstein 1953), and conventions emerge from coordination pressure (Lewis 1969). We measure: (1) the Lewis signaling game with reinforcement (Roth-Erev 1995; Skyrms 2010): agents converge from 0.25 chance to ~0.98 success and transmitted information reaches exactly $\log_2(4) = 2.000$ bits; (2) the genetic code (Nirenberg & Matthaei 1961): 64 codons to 21 meanings, redundancy 3.05, and measured 24.4% of point mutations are synonymous (4.2% nonsense); (3) Shannon (1948): communication is common information. Meaning is the 0/0 - a symbol is nothing in itself and everything in its role; language is empathy scaled to a culture.

1. The Lewis Signaling Game

$I(\text{state}; \text{action}): 0 \text{ bits} \rightarrow \log_2(4) = 2.000 \text{ bits}$

Sender sees a state, emits a symbol; receiver maps it back. Reward only on agreement. Chance performance is $1/4 = 0.25$. Reinforcement drives agents to a convention: measured success 0.984 (seed 42), and the greedy code transmits information $I(\text{state}; \text{action}) = 2.000$ bits in every seed - the theoretical maximum. The sender and receiver codes agree: they now SHARE a language. The symbol itself carries no meaning; the RELATION is the meaning (the 0/0).

2. The Genetic Code (Nirenberg & Matthaei 1961)

64 codons / 21 meanings = 3.05 redundancy

Biology's own language: 64 codons map to 20 amino acids plus STOP - 3.05-fold redundancy. Point mutations measured across 3 positions x 3 alternatives (549 events): 24.4% are synonymous (silent, same protein) and only 4.2% hit a stop codon. The degenerate code is error-tolerant: the language of life resists change while remaining extensible. A codon's meaning is its translation role - nothing intrinsic.

3. Language Is Shared Prediction (Shannon 1948)

A code transmits entropy from one mind to another: communication is common information. Codons carry 5.4 bits; amino acids 4.4 bits. Language is empathy scaled to a whole culture (Ch.60): symbols carry prediction across the self/other boundary. Where two agents exchange bits, two inner worlds share a prior.

4. The 0/0 Proof

The symbol is NOTHING by itself - an empty token. The symbol is EVERYTHING in its role - the shared code. Meaning is the 0/0: private worlds become public at the boundary of two minds. The token is the 0, the use is the x. Etymology, pragmatics, code: all are the meeting point of two information patterns.

5. Connections

Concept	Prior Chapter	Connection
Golden Rule	Ch.60 Golden Rule	Empathy becomes linguistics
Suffering	Ch.59 Problem of Evil	Shared symbols lower surprise

Networks	Ch.45 Complex Networks	<i>Puno (2026) - Meaning: 0/0 of Language</i> Language: network of conventions
The self	Ch.56 The Self	Private mind becomes public
Eternal return	Ch.57 Eternal Return	The message survives the sender
Measurement	Ch.49 Measurement	Observer/observed share a code

6. Conclusion

Language is the 0/0 of meaning: nothing in itself, everything in role. Measured: signaling agents converge on a convention and transmit exactly $\log_2(4) = 2.000$ bits; the genetic code guards its meanings with 3.05-fold redundancy and survives 24.4% of point mutations silently. The symbol is where two minds become one - the boundary of the 0/0. Like the zero itself, a word is empty; like the zero itself, it carries the whole number.

References

- [1] Lewis (1969), *Convention: A Philosophical Study*
- [2] Roth & Erev (1995), *Learning in extensive-form games*
- [3] Skyrms (2010), *Signals: Evolution, Learning, and Information*
- [4] Nirenberg & Matthaei (1961), *The dependence of cell-free protein synthesis*
- [5] Shannon (1948), *A Mathematical Theory of Communication*
- [6] Wittgenstein (1953), *Philosophical Investigations*
- [7] Puno, "The Removable Singularity" (2026)